Module 3

Power System Protection

Why Do We Need the Per-Unit System?

Working directly in ohms, amps and volts across a network of transformers is unwieldy

The Problem

- A power system has generators, transformers, and lines operating at many different voltage levels.
- Impedances in ohms cannot be compared or combined directly unless first referred through every transformer ratio in between.
- Manufacturers rate machines in %R and %X on their own nameplate rating — not a common base.

The Fix — Per-Unit Value

Express every quantity as a fraction of a chosen base value:

$$\text{Per unit value} = \frac{\text{actual value}}{\text{base value (same unit)}}$$

Once every quantity is on the same base, apparatus of any size can be compared, combined, and connected across transformers as one simplified network.

Base Quantities

Four base quantities are linked by Ohm's law — fixing any two fixes the rest

S_B

Base power

kVA or MVA

V_B

Base voltage

kV

I_B

Base current

A

Z_B

Base impedance

Ω (ohm)

Governing Relations (single-phase / per-phase basis)

Only two base quantities need to be chosen independently — usually base kVA and base kV. The other two then follow automatically.

$$S_B = V_B \times I_B$$

$$Z_B = V_B / I_B = (V_B)^2 / S_B$$

Base kVA and base kV are fixed independently. Base Current and base impedance are determined as follows

$$S_B = V_B \times I_B$$

So, $I_B = S_B/V_B$

Base Current

$$I_B = \frac{kVA_B}{kV_B} \quad amperes$$

$$Z_B = V_B / I_B$$

(but $I_B = S_B/V_B$)

$$Z_B = (V_B)^2 / S_B$$

Base Impedance

$$Z_B = \frac{(kV_B)^2 \times 1000}{kVA_B} \quad ohms$$

Base Values — Single-Phase Circuits

Base kVA and base kV (phase value) are chosen; current & impedance follow

Base Current

$$I_B = \frac{kVA_B}{kV_B} \quad \textit{amperes}$$

Base Impedance

$$Z_B = \frac{(kV_B)^2 \times 1000}{kVA_B} \quad \textit{ohms}$$

For single-phase, base kV is the phase (line-to-neutral) voltage and base kVA is the per-phase kVA.

Per-Unit Impedance

$$Z_{pu} = \frac{\text{Actual impedance } (\Omega)}{\text{Base impedance } Z_B}$$

Equivalently:

$$Z_{pu} = \frac{\text{Actual } Z \times kVA_B}{(kV_B)^2 \times 1000}$$

Notice %X and %R are simply p.u. × 100, evaluated at the machine's own rated base.

Base Values — Three-Phase Circuits

Line-to-line kV and total three-phase kVA are used directly as base

Base Current (line current)

$$I_B = \frac{kVA_B}{\sqrt{3} \times kV_B} \quad \textit{amperes}$$

Base Impedance (per phase)

$$Z_B = \frac{(kV_B)^2 \times 1000}{kVA_B} \quad \textit{ohms}$$

Here base kV is the line-to-line voltage and base kVA is the total 3-phase kVA — yet the impedance formula comes out identical in form to the single-phase case.

Where the √3 Goes

Starting from the star-connected phase relations,

$$Z_B = \frac{kV_B / \sqrt{3} \times 1000}{I_B}$$

substituting the line current expression and simplifying cancels the √3, leaving the same compact form as single-phase base impedance.

Because of this, per-unit calculations in power systems are almost always carried out on a 3-phase kVA / line-to-line kV basis, regardless of connection.

Single-Phase vs Three-Phase Base Values — Side by Side

Quantity	Single-Phase	Three-Phase
Base voltage, kV_B	Phase (line-to-neutral) voltage	Line-to-line voltage
Base power, kVA_B	Per-phase kVA	Total 3-phase kVA
Base current, I_B	kVA_B / kV_B	kVA_B / ($\sqrt{3} \times kV_B$)
Base impedance, Z_B	$(kV_B)^2 \times 1000 / kVA_B$	$(kV_B)^2 \times 1000 / kVA_B$ — same form
Per-unit impedance	Actual Z / Z_B	Actual Z / Z_B — same form

Key takeaway: the base-impedance and per-unit-impedance formulas are identical in structure for 1-ϕ and 3-ϕ systems — only the meaning of kV_B and kVA_B changes.

Change of Base

Nameplate p.u. impedance must be converted to the system's common base before use

Equipment is rated in % / p.u. impedance on its OWN base (kVA, kV). A network study picks ONE common base for everything. Convert using:

$$Z_{pu,new} = Z_{pu,old} \times \frac{kVA_{new}}{kVA_{old}} \times \frac{(kV_{old})^2}{(kV_{new})^2}$$

Doubling the base kVA doubles the p.u. impedance on the new base.

$$\left(\frac{Z_{pu,\text{new}}}{Z_{pu,\text{old}}}\right)\left(\frac{V_{\text{new}}}{V_{\text{old}}}\right)^2 = \frac{S_{\text{new}}}{S_{\text{old}}}$$

A transformer's p.u. ipedance is identical whether referred to its primary or secondary side.

$$P = VI$$

$$I = \frac{V}{Z}$$

$$P = V\left(\frac{V}{Z}\right)$$

$$P = \frac{V^2}{Z}$$

$$\boxed{PZ = V^2}$$

$$P_{\text{new}} Z_{\text{new}} = V_{\text{new}}^2$$

$$P_{\text{old}} Z_{\text{old}} = V_{\text{old}}^2$$

Now divide **new by old**:

$$\boxed{\frac{P_{\text{new}}}{P_{\text{old}}} \frac{Z_{\text{new}}}{Z_{\text{old}}} = \left(\frac{V_{\text{new}}}{V_{\text{old}}}\right)^2}$$

$$Z_{pu} = \frac{Z_{\text{actual}}}{Z_{\text{base}}}$$

so the PU ratio is the **inverse of the impedance-base ratio**:

$$\frac{Z_{pu,\text{new}}}{Z_{pu,\text{old}}} = \frac{Z_{B,\text{old}}}{Z_{B,\text{new}}}$$

Hence,

$$\boxed{\frac{Z_{pu,\text{new}}}{Z_{pu,\text{old}}} = \frac{P_{\text{new}}}{P_{\text{old}}} \left(\frac{V_{\text{old}}}{V_{\text{new}}}\right)^2}$$

$$\left(\frac{Z_{pu,\text{new}}}{Z_{pu,\text{old}}}\right) \left(\frac{V_{\text{new}}}{V_{\text{old}}}\right)^2 = \frac{S_{\text{new}}}{S_{\text{old}}}$$

Worked Example

Single-phase transformer — leakage reactance to per-unit

Given

- Single-phase transformer: 20 kVA, 2000/200 V
- Leakage reactance referred to the low-voltage (200 V) side: X = 0.8 Ω
- Find: per-unit leakage reactance

Base values (LV side)

$kV_B = 0.2$ kV, $kVA_B = 20$ kVA

$Z_B = (0.2)^2 \times 1000 / 20 = 2\ \Omega$

Solution

$$Z_{pu} = \frac{\text{Actual } x}{Z_B}$$

$$= 0.8 / 2$$

$$\mathbf{X_{pu} = 0.4\ p.u.}$$

Check on the HV side: $kV_B = 2$ kV, so $Z_B = (2)^2 \times 1000/20 = 200\ \Omega$. Referred to HV, $X = 0.8 \times (2000/200)^2 = 80\ \Omega$, giving p.u. reactance = 80/200 = 0.4 p.u. — identical, confirming p.u. impedance is independent of side.

Base Impedance

$$Z_B = \frac{(kV_B)^2 \times 1000}{kVA_B} \quad \textit{ohms}$$

Base values (LV side)

$kV_B = 0.2$ kV, $kVA_B = 20$ kVA

$Z_B = (0.2)^2 \times 1000 / 20 = 2\ \Omega$

$$Z_{pu} = \frac{\text{Actual } x}{Z_B}$$

$$= 0.8 / 2$$

$$\mathbf{X_{pu} = 0.4\ p.u.}$$

Principle of Referring Impedance Across a Transformer

• Referring an impedance means replacing it with an equivalent impedance on the other side of the transformer.

• The referred impedance produces the same power effect **(Actual and reactive)** when carrying the corresponding current on that side.

For resistance:

$$\text{Original copper loss} = I_1^2 R_1$$

When R_1 is referred to the secondary:

$$I_2^2 R_1' = I_1^2 R_1$$

$$\text{Since } I_2 = aI_1, \text{ where } a = N_1/N_2,$$

$$R_1' = R_1/a^2$$

Same power effect → Different V and I → Impedance changes by a^2

Per unit impedance referred to the HV Side

$$\text{on the HV side: } kV_B = 2\ kV,$$
$$\text{so } Z_B = (2)^2 \times 1000/20 = 200\ \Omega$$

Since we move from the LV side to the HV side, the value of k = 2000/200

X referred to HV = X referred to LV X k^2

$$X = 0.8 \times (2000/200)^2 = 80\ \Omega$$
$$\text{p.u. reactance} = 80/200 = 0.4\ \text{p.u}$$

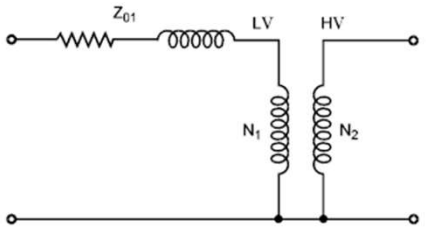
Eliminating the Transformer — Referring the Impedance

The leakage impedance can be referred to either side — whichever matches your chosen base voltage

1. The impedance can be referred either way across the winding

$$Z_{02} = Z_{01} \times (N_2/N_1)^2$$ *primary → secondary*

$$Z_{01} = Z_{02} \times (N_1/N_2)^2$$ *secondary → primary*



2.10. *Simplified Equivalent Circuit of a Single-Phase Two Winding Transformer*

Both equations describe the same physical impedance. Use whichever direction refers the impedance to the side where you have already fixed a base voltage.

2. Convert the referred impedance to per-unit, on that same side's base

$$Z_{pu} = \frac{\text{Referred impedance } (\Omega)}{\text{Base impedance on that same side}}$$

Eliminating the Transformer — Same Result, Either Direction

Once bases follow the transformer ratio, the referral direction no longer matters

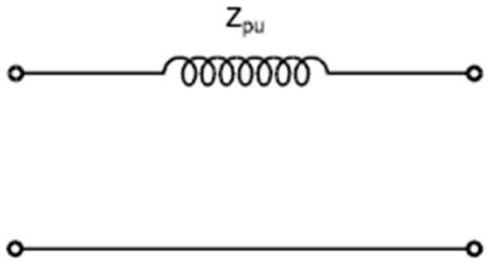
3. Choose base kV in the transformer's own turns ratio

$$kV_{B2} / kV_{B1} = N_2 / N_1$$

Crossing the transformer, scale the base voltage exactly as the transformer itself scales actual voltage — this rule applies the same way whether you referred Z to the primary or the secondary.

4. Either direction gives the same per-unit value

$$Z_{01,pu} = Z_{02,pu} = Z_{pu}$$



Whether you refer the impedance to the primary side or the secondary side, converting to per-unit on the matching base gives the identical number. The turns ratio N_1:N_2 cancels out completely — a single per-unit impedance Z^{pu} represents the transformer, so it can be dropped from the diagram.

1 Simplified calculations

No repeated √3 or transformer-ratio bookkeeping once everything is on one base.

2 Meaningful comparison

Machine parameters (X'', X', X) fall in a narrow, predictable p.u. range regardless of physical rating — erroneous data stand out.

3 Transformers vanish from the diagram

With the correct base change across each winding, ideal transformers disappear from the per-unit impedance diagram entirely.

4 One base serves the whole network

Once chosen, the same base kVA is used system-wide; only base kV changes (by transformation ratio) across each transformer.

Example 2.6. Determine the per unit impedance of a transmission line having an impedance of $(30 + j\,110)\ \Omega$ on 100 MVA and 132 kV base voltage.

Solution : Impedance of line, $Z = (30 + j\,110)\ \Omega$

Base kVA, $kVA_B = 100$ MVA or 100,000 kVA

Base kV, $kV_B = 132$ kV

$$\text{Per unit impedance, } Z_{pu} = \frac{Z \times kVA_B}{(kV_B)^2 \times 1{,}000}$$

$$= \frac{(30 + j110) \times 100{,}000}{(132)^2 \times 1{,}000}$$

$$= (0.172 + j\,0.631)\ \text{pu}$$ **Ans.**

Example 2.7. A 30 MVA, 11 kV generator has a reactance of 0.2 pu referred to its ratings as bases. Determine the per unit reactance when referred to base kVA of 50,000 kVA and base kV of 33 kV.

Solution : Old base kVA = 30,000 kVA

Old base kV = 11 kV

Old per unit impedance,

$$Z_{pu\ old} = 0.2 \text{ pu}$$

New base kVA = 50,000 kVA

New base kV = 33 kV

New per unit impedance,

$$Z_{pu\ new} = Z_{pu\ old} \times \frac{\text{New base kVA}}{\text{Old base kVA}} \times \left(\frac{\text{Old base kV}}{\text{New base kV}} \right)^2$$

$$= 0.2 \times \frac{50{,}000}{30{,}000} \times \left(\frac{11}{33} \right)^2 = 0.037 \text{ pu } \textbf{Ans.}$$

A balanced **3-phase system** has:

$$S = 30\ \text{MVA}, \qquad V_L = 12\ \text{kV}, \qquad Z = 1.5 + j2.0\ \Omega/\text{phase}$$

Choose the initial bases as:

$$S_{B1} = 50\ \text{MVA}, \qquad V_{B1} = 10\ \text{kV}$$

Find:

1. Base current and base impedance
2. Per-unit voltage, current and impedance
3. The per-unit impedance on a new base of **100 MVA, 20 kV**

1. Initial base quantities

For a 3-phase system,

$$I_{B1} = \frac{S_{B1}}{\sqrt{3}V_{B1}} = \frac{50}{\sqrt{3}(10)}$$

$$\boxed{I_{B1} = 2.887\ \text{kA}}$$

Base impedance:

$$Z_{B1} = \frac{V_{B1}^2}{S_{B1}} = \frac{10^2}{50}$$

$$\boxed{Z_{B1} = 2\ \Omega}$$

Voltage

$$V_{pu} = \frac{V}{V_{B1}} = \frac{12}{10}$$

$$\boxed{V_{pu} = 1.2}$$

Current

Actual line current is

$$I = \frac{S}{\sqrt{3}V} = \frac{30}{\sqrt{3}(12)} = 1.443 \text{ kA}$$

Therefore,

$$I_{pu} = \frac{1.443}{2.887}$$

$$\boxed{I_{pu} = 0.5}$$

Impedance

$$Z_{pu,1} = \frac{Z}{Z_{B1}} = \frac{1.5 + j2.0}{2}$$

$$\boxed{Z_{pu,1} = 0.75 + j1.0 \, pu}$$

Notice also that

$$|Z| = \sqrt{1.5^2 + 2^2} = 2.5\Omega$$

so

$$|Z|_{pu} = \frac{2.5}{2} = 1.25 \, pu$$

3. Change the base

Now choose:

$$S_{B2} = 100\ \text{MVA}, \qquad V_{B2} = 20\ \text{kV}$$

Use

$$Z_{pu,2} = Z_{pu,1} \left(\frac{S_{B2}}{S_{B1}}\right) \left(\frac{V_{B1}}{V_{B2}}\right)^2$$

Substitute:

$$Z_{pu,2} = (0.75 + j1.0) \left(\frac{100}{50}\right) \left(\frac{10}{20}\right)^2$$

$$= (0.75 + j1.0)(2)(0.25)$$

Therefore,

$$\boxed{Z_{pu,2} = 0.375 + j0.50\ pu}$$

Symmetrical Components & Sequence Networks

From unbalanced faults to solvable circuits — a step-by-step visual guide to one of power engineering's most powerful analytical tools.

The Core Trick: Three "Balanced Worlds" Inside One Unbalanced System

Any unbalanced 3-phase set can be mathematically decomposed into three perfectly balanced sequences. Think of it like a prism splitting white light into three pure colours — each sequence travels independently and recombines to recreate the original messy signal.

Positive Sequence

Balanced, ABC rotation. The "normal" operating world. Like three friends walking in step.

Negative Sequence

Balanced, ACB rotation — spinning backwards. Indicates asymmetric disturbance or fault.

Zero Sequence

All three phases in phase with each other. The "earth-return" world — only exists when current flows to ground.

Key principle: **Va = V1 + V2 + V0** — every phase quantity is the superposition of three sequence components.

Introduction to Symmetrical Components

An **unbalanced 3-phase system** can be represented as the sum of **three balanced sets of components**:

$$\boxed{\text{Unbalanced 3-}\phi = \text{Positive Sequence} + \text{Negative Sequence} + \text{Zero Sequence}}$$

Positive Sequence (1)

- Three equal phasors, 120° apart
- Normal phase order: **a-b-c**
- Represents the **normal balanced component**

Negative Sequence (2)

- Three equal phasors, 120° apart
- Reverse phase order: **a-c-b**
- Appears due to **system unbalance**

Zero Sequence & Application to Fault Analysis

Zero Sequence (0)

- Three equal phasors
- All three phasors are **in phase**

$$\boxed{V_{a0} = V_{b0} = V_{c0}}$$

- Particularly important in **faults involving ground**

Combining the Sequence Components

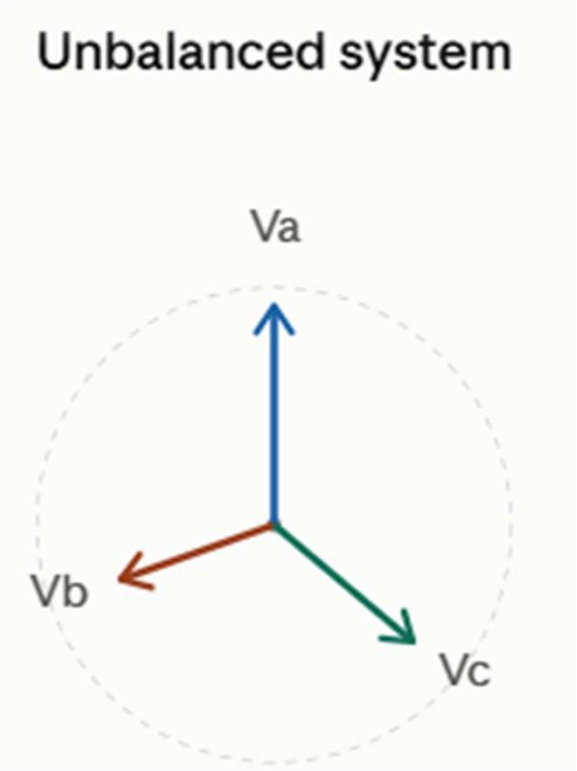
$$\boxed{V_a = V_{a1} + V_{a2} + V_{a0}}$$

Similarly for phases b and c.

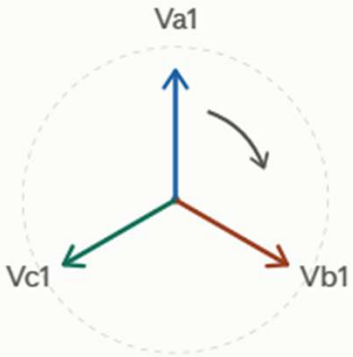
Key Idea

$$\boxed{\text{Unbalanced Fault} \rightarrow \text{Sequence Networks} \rightarrow \text{Simpler Analysis}}$$

Symmetrical components allow unbalanced faults to be analyzed using balanced sequence networks.

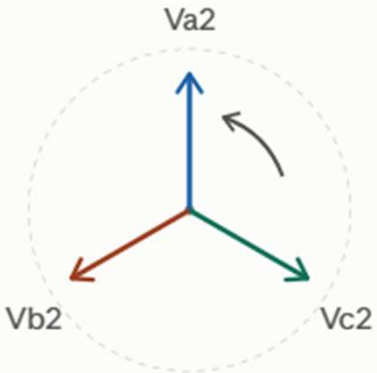

Unbalanced system
Va
Vb
Vc

Positive sequence



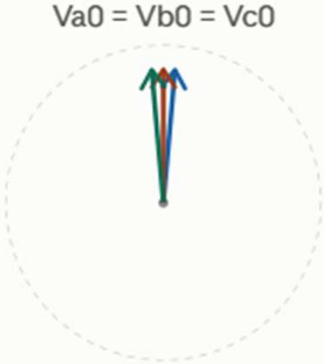
Rotates a-b-c, same order as the original system

Negative sequence



Rotates a-c-b, reversed order

Zero sequence



Equal magnitude, in phase, no rotation

Sequence Components Under Balanced Conditions

Balanced 3-Phase System

For a perfectly balanced three-phase system:

$$V_a = V\angle 0^\circ, \qquad V_b = V\angle -120^\circ, \qquad V_c = V\angle 120^\circ$$

Sequence Components

Positive Sequence (V_1) — EXISTS

- Equal magnitudes
- 120° apart
- Normal phase sequence: **a-b-c**

Negative Sequence (V_2) — ZERO

- Appears when the system is unbalanced

Zero Sequence (V_0) — ZERO

- Appears under certain unbalanced conditions, particularly ground faults

$$\boxed{V_1 = V, \qquad V_2 = 0, \qquad V_0 = 0}$$

Sequence Components in Fault Analysis

Balanced 3-Phase Fault

A three-phase fault is a **balanced fault**.

Therefore,

$$\boxed{\text{Balanced 3-Phase Fault} \Rightarrow \text{Positive-Sequence Network Only}}$$

Hence,

$$\boxed{V_2 = 0, \qquad V_0 = 0}$$

Unbalanced Faults

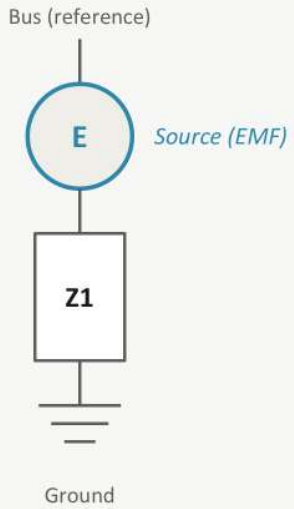
Negative- and/or zero-sequence components appear during **unbalanced faults**.

- **Line-to-Line (L-L) fault:** Positive + Negative sequence
- **Single Line-to-Ground (L-G) fault:** Positive + Negative + Zero sequence
- **Double Line-to-Ground (L-L-G) fault:** Positive + Negative + Zero sequence

Key Point

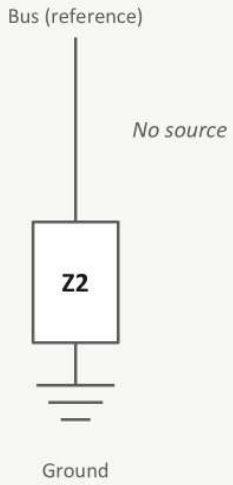
$$\boxed{\text{Negative and Zero Sequences} \Rightarrow \text{Used for Unbalanced Fault Analysis}}$$

Positive-sequence network



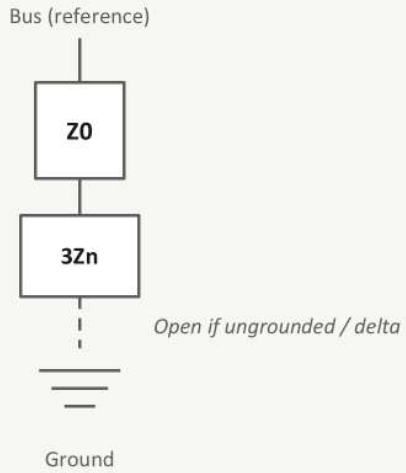
- Looks like the normal single-line diagram used for load flow
- Includes EMF sources — generator internal voltages drive this network
- Impedances: generator reactance, transformer leakage reactance, line impedance
- Neutral / ground is the reference bus
- The only network with sources under balanced operation

Negative-sequence network



- Same static impedances as positive sequence — lines and transformers don't care about phase rotation
- No EMF sources — machines don't generate negative-sequence voltage internally
- Rotating machines use X2, usually close to the subtransient reactance
- Reference bus is grounded but carries no source voltage

0 Zero-sequence network



- Topology depends on grounding and transformer connections
- Needs a physical path through neutral to flow
- Grounded-Y: path exists — model 3×Zn in series with neutral
- Ungrounded-Y or delta: path is open — current can't escape that side
- No EMF sources

Actual Phase Quantities → Sequence Components

Given the actual phase quantities $V_a,\ V_b,\ V_c$:

$$\boxed{\begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix}}$$

Therefore,

$$V_0 = \frac{1}{3}(V_a + V_b + V_c)$$

$$V_1 = \frac{1}{3}(V_a + aV_b + a^2V_c)$$

$$V_2 = \frac{1}{3}(V_a + a^2V_b + aV_c)$$

where

$$\boxed{a = 1\angle 120^\circ, \qquad a^2 = 1\angle 240^\circ, \qquad 1 + a + a^2 = 0}$$

Sequence Components → Actual Phase Quantities

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix}$$

Hence,

$$V_a = V_0 + V_1 + V_2$$

$$V_b = V_0 + a^2 V_1 + a V_2$$

$$V_c = V_0 + a V_1 + a^2 V_2$$

The same transformation applies to currents: simply replace V by I

Approach to Fault Problems Using Sequence Networks

1. Convert All Impedances to a Common PU Base

Determine:

$$\boxed{Z_1, \qquad Z_2, \qquad Z_0}$$

2. Identify the Type of Fault

Fault	Sequence Networks Required
3-phase fault	Positive only
L-L fault	Positive + Negative
L-G fault	Positive + Negative + Zero
L-L-G fault	Positive + Negative + Zero

Overall Problem-Solving Flow

One-Line Diagram

↓

Select Common Base

↓

Convert All Data to PU

↓

Construct $(+), (-), (0)$ Sequence Networks

↓

Reduce Networks to Fault Point

↓

Connect Networks According to Fault

↓

Calculate I_0, I_1, I_2

↓

Find Actual Phase Currents / Voltages

The phase voltages are

$$V_a = 100\angle 0^\circ, \qquad V_b = 100\angle -120^\circ, \qquad V_c = 100\angle 120^\circ \text{ V}$$

Find the symmetrical components.

apply the transformation matrix:

$$\begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} 100 \\ 100a^2 \\ 100a \end{bmatrix}$$

Multiplying,

$$= \frac{100}{3} \begin{bmatrix} 1 + a + a^2 \\ 1 + a^3 + a^3 \\ 1 + a^4 + a^2 \end{bmatrix}$$

Using

$$a^3 = 1, \qquad 1 + a + a^2 = 0$$

we get

$$\begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 100 \\ 0 \end{bmatrix}$$

Therefore,

$$\boxed{V_0 = 0, \qquad V_1 = 100\angle 0^\circ, \qquad V_2 = 0}$$

Given

$$V_a = 100\angle 0^\circ, \qquad V_b = 100\angle 120^\circ, \qquad V_c = 100\angle -120^\circ \text{ V}$$

Find the symmetrical components.

Write

$$V_b = 100a, \qquad V_c = 100a^2$$

Then

$$\begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} 100 \\ 100a \\ 100a^2 \end{bmatrix}$$

which gives

$$\boxed{\begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 100 \end{bmatrix}}$$

Hence,

$$\boxed{V_0 = 0, \qquad V_1 = 0, \qquad V_2 = 100\angle 0^\circ}$$

Given

$$V_a = V_b = V_c = 60\angle 0^\circ \text{ V}$$

Find the symmetrical components.

$$\begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} 60 \\ 60 \\ 60 \end{bmatrix}$$

Therefore,

$$= \frac{60}{3} \begin{bmatrix} 3 \\ 1 + a + a^2 \\ 1 + a + a^2 \end{bmatrix}$$

Hence,

$$\boxed{\begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 60 \\ 0 \\ 0 \end{bmatrix}}$$

Therefore,

$$\boxed{V_0 = 60\angle 0^\circ, \qquad V_1 = 0, \qquad V_2 = 0}$$

$$V_a = 100\angle 0^\circ \text{ V}$$

$$V_b = 80\angle -120^\circ \text{ V}$$

$$V_c = 60\angle 120^\circ \text{ V}$$

Find $V_0,\ V_1,\ V_2$.

Write

$$V_a = 100, \qquad V_b = 80a^2, \qquad V_c = 60a$$

Use

$$\begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} 100 \\ 80a^2 \\ 60a \end{bmatrix}$$

This gives

$$V_0 = \frac{1}{3}(100 + 80a^2 + 60a)$$

$$V_1 = \frac{1}{3}(100 + 80 + 60)$$

$$V_2 = \frac{1}{3}(100 + 80a + 60a^2)$$

Therefore,

$$\boxed{V_1 = 80\angle 0^\circ \text{ V}}$$

and evaluating the other two,

$$\boxed{V_0 = 11.55\angle -30^\circ \text{ V}}$$

$$\boxed{V_2 = 11.55\angle 30^\circ \text{ V}}$$

So,

$$\boxed{V_0 = 11.55\angle -30^\circ, \quad V_1 = 80\angle 0^\circ, \quad V_2 = 11.55\angle 30^\circ}$$

Given

$$I_0 = 2\angle 0^\circ, \qquad I_1 = 10\angle 0^\circ, \qquad I_2 = 4\angle 0^\circ \text{ A}$$

find I_a, I_b, I_c.

Use:

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} 2 \\ 10 \\ 4 \end{bmatrix}$$

Multiplying,

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 2 + 10 + 4 \\ 2 + 10a^2 + 4a \\ 2 + 10a + 4a^2 \end{bmatrix}$$

Using

$$a = -0.5 + j0.866, \qquad a^2 = -0.5 - j0.866$$

gives

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 16 \\ -5 - j5.196 \\ -5 + j5.196 \end{bmatrix}$$

$$\boxed{I_a = 16\angle 0^\circ \text{ A}}$$

$$\boxed{I_b = 7.21\angle -133.9^\circ \text{ A}}$$

$$\boxed{I_c = 7.21\angle 133.9^\circ \text{ A}}$$

Conclusion: The Grid's Symmetrical Backbone

From slide-rule calculations to millisecond digital protection — symmetrical components remain the universal language of power system fault analysis.

The Core Insight

Any unbalanced fault, no matter how complex, reduces to three independent, solvable single-phase circuits — connected by precise mathematical rules at one point.

The Engineering Value

Sequence networks underpin every protective relay setting, every fault-level calculation, and every stability study on the modern grid.

The Future Challenge

As inverter-based renewables dominate generation, their different fault signatures challenge traditional sequence-based methods — demanding new understanding whilst building on these same foundations.

Master the three sequences and you hold the key to understanding every fault, every relay, and every stability limit on the power network.

Power System Fault Analysis

Introduction to Faults in Power Systems

A fault is an abnormal condition in a power system that causes excessive current and/or abnormal voltages.

Common Causes

- Lightning and insulation failure
- Equipment failure or ageing
- Trees, animals, or foreign objects contacting conductors
- Mechanical damage and human error

Why Fault Analysis is Important

Faults can cause:

$$\boxed{\text{High Fault Current} \rightarrow \text{Equipment Damage} \rightarrow \text{System Disturbance}}$$

Fault analysis is required to:

- Determine the magnitude of fault current
- Select appropriate circuit breakers and protective relays
- Ensure safe and reliable operation of the power system

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Basic Classification

$$\boxed{\text{Power-System Faults} \rightarrow \begin{cases} \text{Symmetrical Faults} \\ \text{Unsymmetrical Faults} \end{cases}}$$

Symmetrical: $L-L-L,\ L-L-L-G$

Unsymmetrical: $L-G,\ L-L,\ L-L-G$

Defining the Shunt Fault

A shunt fault is an **abnormal low-resistance path** that bypasses the normal load circuit — arising from direct physical contact between conductors, or between a conductor and earth.

Low-Resistance Path

Unintended connection dramatically reduces impedance at the fault point

Extreme Current Surge

Current rises far beyond rated values, stressing all connected equipment

Physical Contact

Conductor–conductor or conductor–ground contact triggers the fault



Fault Classification

Symmetrical Faults

Three-Phase (LLL)

All three phases short together — balanced, analysed via single-phase equivalent

Three-Phase-to-Ground (LLLG)

All phases shorted to ground — most severe, positive-sequence network only

Unsymmetrical Faults

Single Line-to-Ground (LG)

Most common — sequence networks connected in series

Line-to-Line (LL)

Two phases contact — Z1 and Z2 networks in parallel

Double Line-to-Ground (LLG)

Two phases and ground — Z1 in series with (Z0 ‖ Z2)

Fault Calculations for Shunt Faults – Overall Procedure

Fault calculations are performed to determine the current flowing during a short-circuit condition.

The general procedure is:

1. Draw the one-line diagram of the power system.
2. Identify the fault location and type of fault.
3. Select common base MVA and base voltage.
4. Convert all system impedances to a common per-unit base.
5. Construct the required sequence networks.
6. Reduce the sequence networks to the fault point.
7. Connect the sequence networks according to the fault type.
8. Calculate sequence currents.
9. Convert sequence currents to phase currents.
10. Convert per-unit values to actual values, if required.

$$\boxed{\text{System Data} \rightarrow \text{PU System} \rightarrow \text{Sequence Networks} \rightarrow \text{Fault Current}}$$

Step 1 – One-Line Diagram and Common Base

Draw the one-line diagram showing:

- Generators
- Transformers
- Transmission lines
- Buses
- Fault location

Select:

$$S_B = \text{Common Base MVA}$$

and choose the base voltage at one bus.

Base voltages at other buses are determined using transformer voltage ratios.

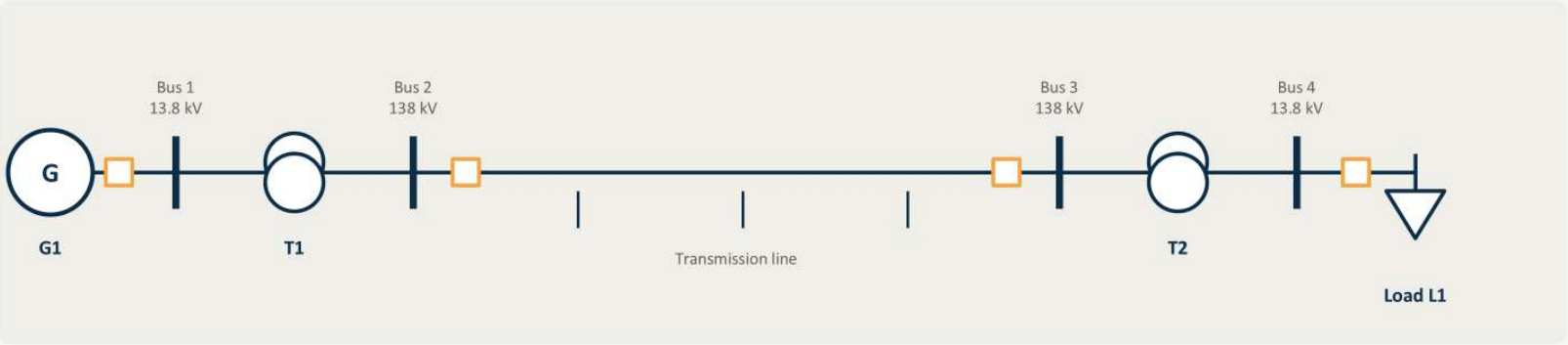
Convert all impedances to the selected base:

$$Z_{pu,new} = Z_{pu,old} \left(\frac{S_{B,new}}{S_{B,old}} \right) \left(\frac{V_{B,old}}{V_{B,new}} \right)^2$$

After this step, all system quantities are on a common per-unit base.

What is a one-line diagram?

A power system has three phases, but they're normally balanced and identical — so engineers draw just one phase as a single line, with a symbol standing in for each piece of equipment.



It's the map every fault, protection, and load-flow study starts from.

The symbols



Generator

Bus



Transformer



Circuit breaker



Transmission line



Load

Step 2 – Construct Sequence Networks

Construct the three sequence networks of the system:

Positive-sequence network

$$Z_1$$

- Represents normal phase sequence
- Contains the generator/source voltage

Negative-sequence network

$$Z_2$$

- Represents reverse phase sequence
- Normally contains no voltage source

Zero-sequence network

$$Z_0$$

- Represents in-phase components
- Depends strongly on transformer connections and grounding

Reduce each network as viewed from the fault point:

$$\boxed{Z_{1,eq}, \quad Z_{2,eq}, \quad Z_{0,eq}}$$

How Do Negative- and Zero-Sequence Currents Flow Without Sources?

A balanced generator produces only positive-sequence EMF:

$$\boxed{E_1 \neq 0, \qquad E_2 = 0, \qquad E_0 = 0}$$

During an unbalanced fault, the fault condition connects the sequence networks together.

For example, for a single line-to-ground (L-G) fault:

$$\boxed{E_1 \rightarrow Z_1 \rightarrow Z_2 \rightarrow Z_0}$$

The positive-sequence source E_1 drives current through all three sequence networks:

$$I_1 = I_2 = I_0 = \frac{E_1}{Z_1 + Z_2 + Z_0}$$

Therefore,

$$\boxed{E_2 = 0,\ E_0 = 0 \quad \text{but} \quad I_2 \neq 0,\ I_0 \neq 0}$$

The negative- and zero-sequence currents produce voltage drops across their respective impedances:

$$V_2 = -I_2 Z_2, \qquad V_0 = -I_0 Z_0$$

Step 3 – Connect Sequence Networks According to Fault Type

The sequence networks required depend on the type of shunt fault.

Fault Type	Sequence Networks
Three-phase fault (L-L-L)	Positive only
Single line-to-ground (L-G)	Positive + Negative + Zero
Line-to-line (L-L)	Positive + Negative
Double line-to-ground (L-L-G)	Positive + Negative + Zero

For example, for an L-G fault:

$$\boxed{I_0 = I_1 = I_2}$$

and the three sequence networks are connected in series.

For an L-L fault:

$$\boxed{I_0 = 0}$$

so the zero-sequence network is not involved.

Why Are All Sequence Currents Equal in an L-G Fault?

Consider a single line-to-ground fault on phase a.

At the fault point:

$$I_a = I_f, \qquad I_b = 0, \qquad I_c = 0$$

Use the phase-to-sequence transformation matrix:

$$\begin{bmatrix} I_0 \\ I_1 \\ I_2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} I_f \\ 0 \\ 0 \end{bmatrix}$$

Multiplying,

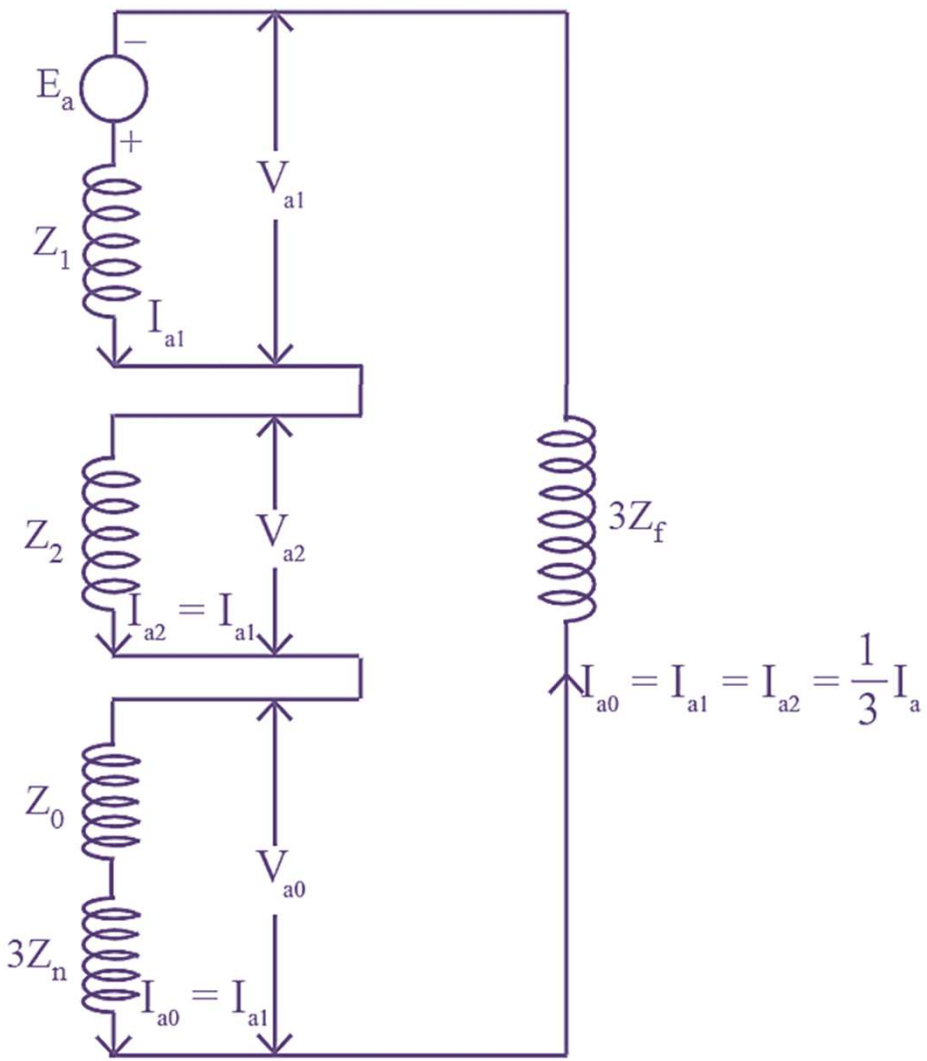
$$\begin{bmatrix} I_0 \\ I_1 \\ I_2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} I_f \\ I_f \\ I_f \end{bmatrix}$$

Therefore,

$$\boxed{I_0 = I_1 = I_2 = \frac{I_f}{3}}$$

Hence, the same current must flow through all three sequence networks.

$$\boxed{I_0 = I_1 = I_2 \;\Rightarrow\; \text{Positive, Negative and Zero Sequence Networks in Series}}$$


E_a
−
+
Z_1
I_a1
V_a1
Z_2
I_a2 = I_a1
V_a2
3Z_f
I_a0 = I_a1 = I_a2 = 1/3 I_a
Z_0
V_a0
3Z_n
I_a0 = I_a1

Step 4 – Calculate Sequence Currents

After connecting the sequence networks, calculate:

$$I_0, \qquad I_1, \qquad I_2$$

The equations depend on the type of fault.

For example, for a solid L-G fault:

$$I_0 = I_1 = I_2 = \frac{E}{Z_1 + Z_2 + Z_0}$$

Hence the faulted phase current is:

$$I_a = I_0 + I_1 + I_2$$

$$\boxed{I_a = \frac{3E}{Z_1 + Z_2 + Z_0}}$$

Usually,

$$E \approx 1\angle 0^\circ \; pu$$

for basic fault calculations.

Step 5 – Convert Sequence Currents to Phase Currents

Once I_0, I_1, I_2 are known, obtain the actual phase quantities using the transformation matrix:

$$\boxed{\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} I_0 \\ I_1 \\ I_2 \end{bmatrix}}$$

where

$$a = 1\angle 120^\circ$$

This gives the fault currents in phases a, b, c.

Step 6 – Convert PU Current to Actual Current

If the calculated current is in per unit, determine the base current:

$$I_B = \frac{S_B}{\sqrt{3} V_B}$$

Then,

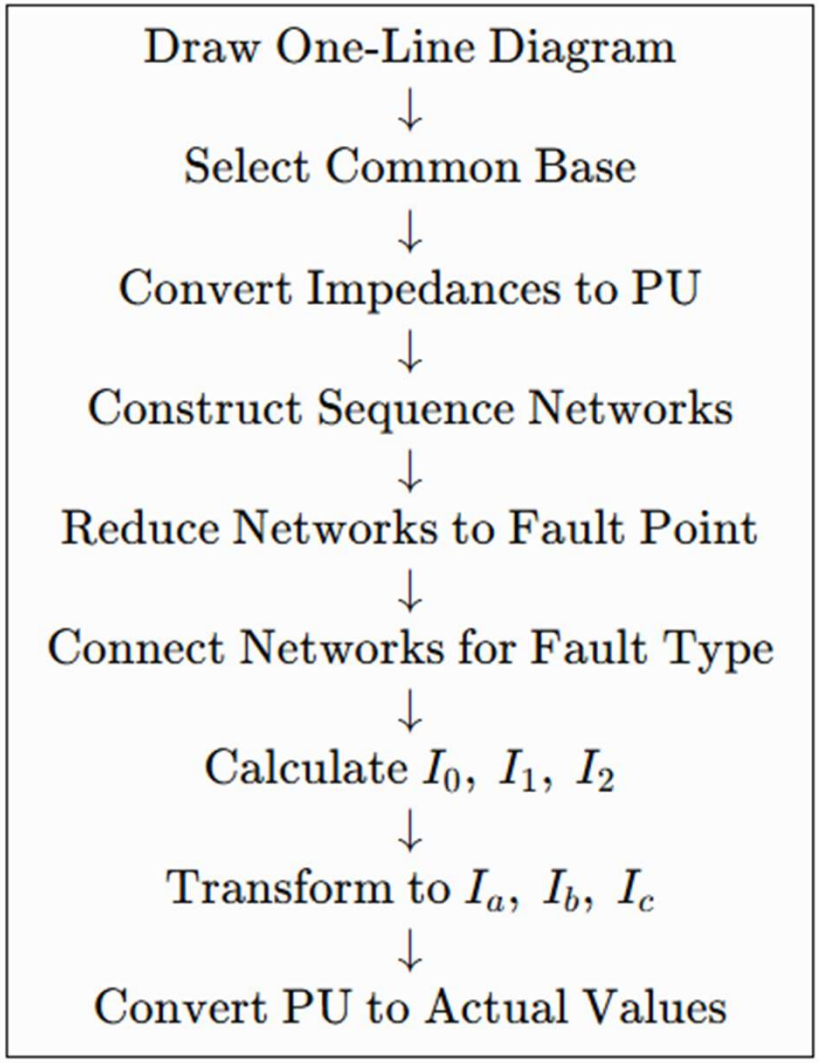
$$\boxed{I_{\text{actual}} = I_{pu} \times I_B}$$

For S_B in MVA and V_B in kV:

$$\boxed{I_B(\text{kA}) = \frac{S_B(\text{MVA})}{\sqrt{3}\, V_B(\text{kV})}}$$

Final results may include:

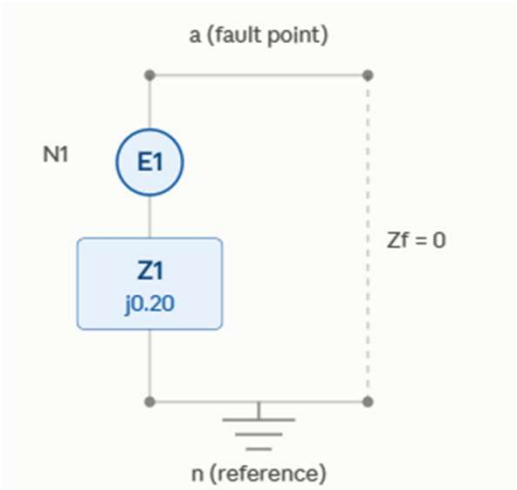
- Fault current in each phase
- Fault current in kA
- Sequence currents
- Bus voltages during the fault


Draw One-Line Diagram
Select Common Base
Convert Impedances to PU
Construct Sequence Networks
Reduce Networks to Fault Point
Connect Networks for Fault Type
Calculate $I_0,\ I_1,\ I_2$
Transform to $I_a,\ I_b,\ I_c$
Convert PU to Actual Values

A generator supplies a bus through a system having positive-sequence equivalent impedance

$$Z_1 = j0.20\ pu$$

A solid three-phase fault occurs at the bus. Determine the fault current.



For a symmetrical three-phase fault, only the positive-sequence network is involved.

$$I_1 = \frac{E_1}{Z_1}$$

Therefore,

$$I_1 = \frac{1}{j0.20}$$

Since

$$\frac{1}{j} = -j$$

$$I_1 = -j5 = 5\angle -90^\circ \; pu$$

For a balanced three-phase fault:

$$I_0 = 0, \qquad I_2 = 0$$

For a balanced three-phase fault:

$$I_0 = 0, \qquad I_2 = 0$$

Using the sequence-to-phase transformation:

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} 0 \\ 5\angle -90^\circ \\ 0 \end{bmatrix}$$

Hence,

$$\boxed{I_a = 5\angle -90^\circ \; pu}$$

$$\boxed{I_b = 5\angle 150^\circ \; pu}$$

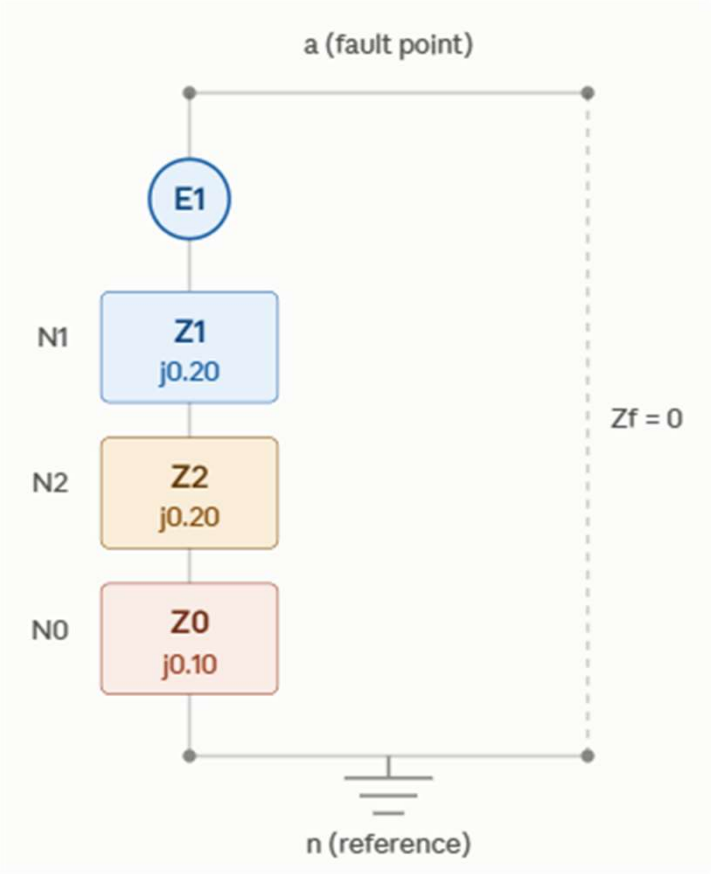
$$\boxed{I_c = 5\angle 30^\circ \; pu}$$

The three fault currents are equal in magnitude and 120° apart.

A three-phase fault occurs at an 11 kV bus. The system base is 100 MVA and the positive-sequence equivalent impedance is

$$Z_1 = j0.25\ pu$$

Determine the fault current in kA.



Fault current:

$$I_f = \frac{1}{j0.25}$$

$$\boxed{I_f = 4\angle -90^\circ \; pu}$$

Base current:

$$I_B = \frac{S_B}{\sqrt{3}V_B}$$

$$I_B = \frac{100}{\sqrt{3}(11)}$$

$$I_B = 5.249 \text{ kA}$$

Therefore,

$$I_{f,\text{actual}} = I_{f,pu} I_B$$

$$= 4(5.249)$$

$$\boxed{|I_f| = 21.0 \text{ kA}}$$

At a particular bus,

$$Z_1 = j0.20, \qquad Z_2 = j0.20, \qquad Z_0 = j0.10\ pu$$

A solid L-G fault occurs on phase a. Determine the sequence currents and phase currents.

Also determine the sequence currents and phase currents if there was a fault impedance Z_f =j0.05 pu

For an L-G fault:

$$I_0 = I_1 = I_2$$

The three sequence networks are connected in series.

Therefore,

$$I_1 = \frac{E_1}{Z_1 + Z_2 + Z_0}$$

$$I_1 = \frac{1}{j(0.20 + 0.20 + 0.10)}$$

$$I_1 = \frac{1}{j0.50} = -j2$$

Hence,

$$\boxed{I_0 = I_1 = I_2 = -j2\ pu}$$

Now use the transformation matrix:

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} -j2 \\ -j2 \\ -j2 \end{bmatrix}$$

Therefore,

$$I_a = -j6$$

and because $1 + a + a^2 = 0$,

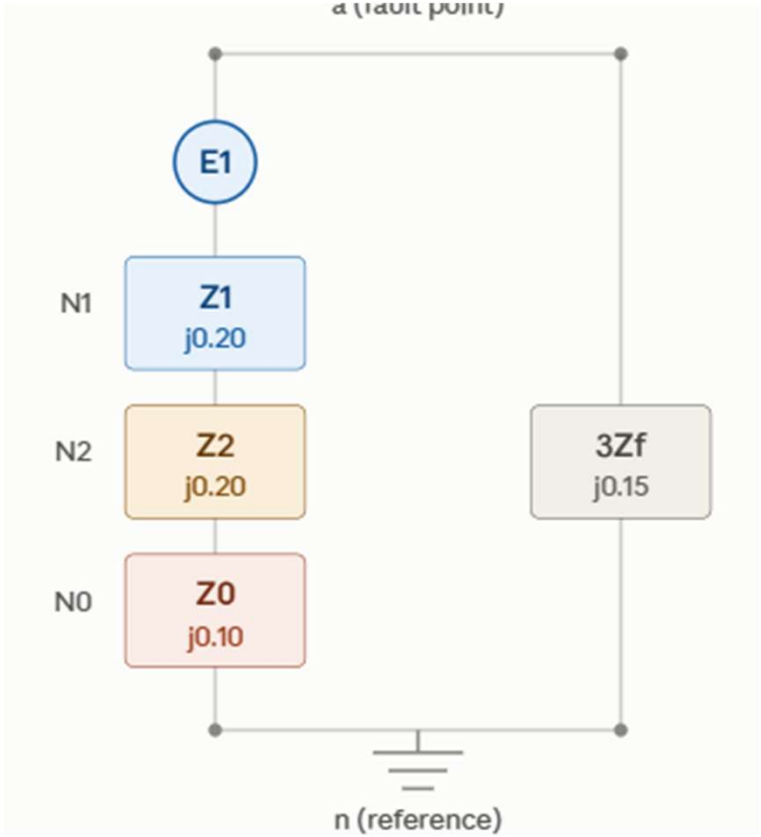
$$I_b = I_c = 0$$

Thus,

$$\boxed{I_a = 6\angle -90^\circ\ pu}$$

$$\boxed{I_b = I_c = 0}$$

This also verifies the physical condition of an L-G fault on phase a.



The fault impedance appears as $3Z_f$, therefore:

$$I_1 = \frac{E_1}{Z_1 + Z_2 + Z_0 + 3Z_f}$$

Substitute:

$$I_1 = \frac{1}{j0.20 + j0.20 + j0.10 + 3(j0.05)}$$

$$I_1 = \frac{1}{j0.65}$$

$$\boxed{I_1 = -j1.538\ pu}$$

Hence,

$$\boxed{I_0 = I_1 = I_2 = -j1.538\ pu}$$

The faulted phase current is:

$$I_a = I_0 + I_1 + I_2$$

$$I_a = 3(-j1.538)$$

$$\boxed{I_a = -j4.615\ pu}$$

or

$$\boxed{I_a = 4.615\angle -90^\circ\ pu}$$

A generator supplies a transmission line through a transformer. A solid line-to-line fault occurs between phases b and c at the receiving-end bus.

The sequence reactances on a common per-unit base are:

Generator:

$$X_{1G} = 0.15\ pu, \qquad X_{2G} = 0.20\ pu$$

Transformer:

$$X_{1T} = X_{2T} = 0.10\ pu$$

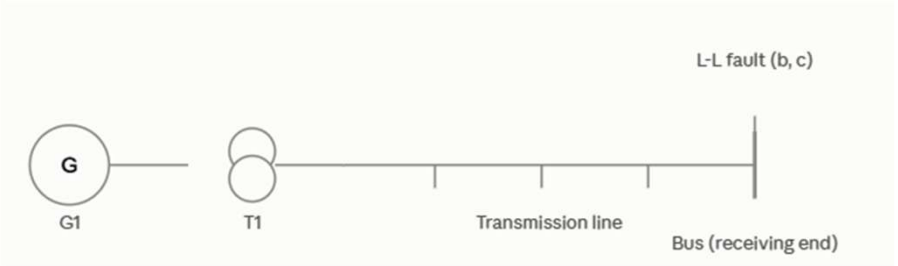
Transmission line:

$$X_{1L} = X_{2L} = 0.05\ pu$$

Assume

$$E_1 = 1\angle 0^\circ\ pu$$

Determine the sequence currents and phase fault currents.



Phase a is not involved in the fault:

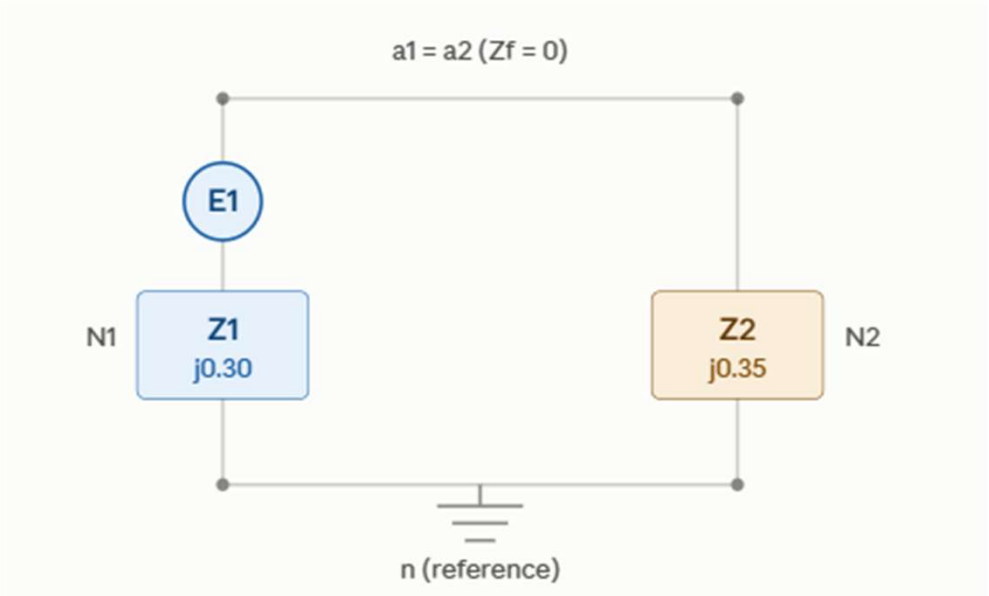
$$\boxed{I_a = 0}$$

There is no connection to ground, so the currents in the two faulted phases are equal and opposite:

$$\boxed{I_b = -I_c}$$

Also, because phases b and c are short-circuited together:

$$\boxed{V_b = V_c}$$



For a line-to-line fault:

$$I_0 = 0$$

and

$$I_2 = -I_1$$

First determine the positive-sequence equivalent impedance:

$$Z_1 = j(X_{1G} + X_{1T} + X_{1L})$$

$$Z_1 = j(0.15 + 0.10 + 0.05)$$

$$\boxed{Z_1 = j0.30\ pu}$$

Now determine the negative-sequence equivalent impedance:

$$Z_2 = j(X_{2G} + X_{2T} + X_{2L})$$

$$Z_2 = j(0.20 + 0.10 + 0.05)$$

$$\boxed{Z_2 = j0.35\ pu}$$

For a solid L-L fault,

$$I_1 = \frac{E_1}{Z_1 + Z_2}$$

Therefore,

$$I_1 = \frac{1}{j0.30 + j0.35}$$

$$I_1 = \frac{1}{j0.65}$$

$$\boxed{I_1 = -j1.538\ pu}$$

Hence,

$$\boxed{I_2 = j1.538\ pu}$$

and

$$\boxed{I_0 = 0}$$

Now obtain the phase currents using the transformation matrix:

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} 0 \\ -j1.538 \\ j1.538 \end{bmatrix}$$

For phase a,

$$I_a = I_1 + I_2 = 0$$

$$\boxed{I_a = 0}$$

For phase b,

$$I_b = a^2 I_1 + a I_2$$

which gives

$$\boxed{I_b = -2.664\ pu}$$

For phase c,

$$I_c = a I_1 + a^2 I_2$$

which gives

$$\boxed{I_c = 2.664\ pu}$$

Therefore,

$$\boxed{I_a = 0, \qquad I_b = -2.664\ pu, \qquad I_c = 2.664\ pu}$$

As expected for a line-to-line fault,

$$\boxed{I_b = -I_c}$$

and no zero-sequence current flows:

$$\boxed{I_0 = 0}$$

At a bus,

$$Z_1 = j0.20, \qquad Z_2 = j0.20, \qquad Z_0 = j0.10\ pu$$

A solid L-L-G fault occurs on phases b and c.

Determine I_0, I_1, I_2.

At the fault point,

$$V_b = 0, \qquad V_c = 0$$

while phase a is not faulted:

$$I_a = 0$$

Now use the sequence-to-phase voltage transformation:

$$V_b = V_0 + a^2 V_1 + a V_2 = 0$$

$$V_c = V_0 + a V_1 + a^2 V_2 = 0$$

Subtracting the two equations,

$$(a^2 - a)(V_1 - V_2) = 0$$

therefore,

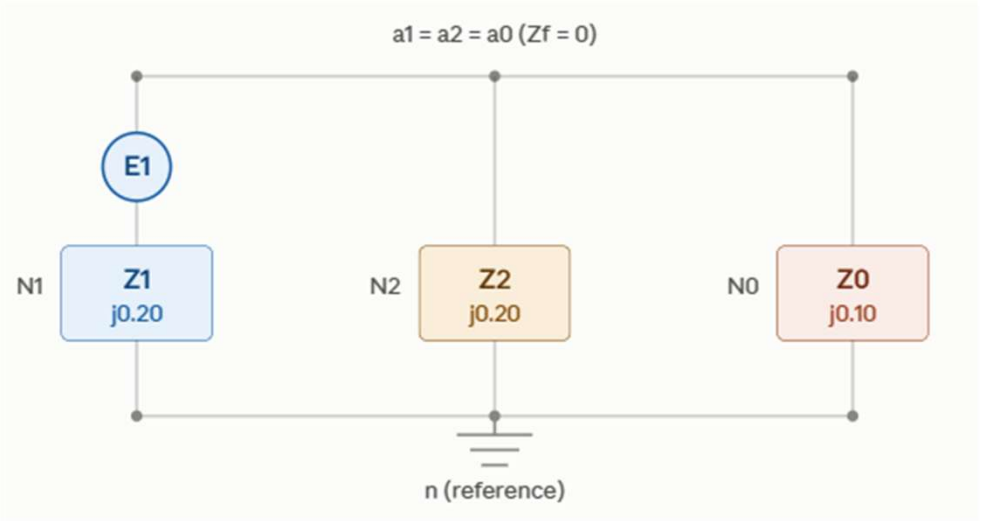
$$\boxed{V_1 = V_2}$$

Substituting back gives

$$\boxed{V_0 = V_1 = V_2}$$

So at the fault point,

$$\boxed{V_1 = V_2 = V_0}$$



For an L-L-G fault, all three sequence networks are required.

The negative- and zero-sequence impedances appear in parallel:

$$Z_2 \parallel Z_0 = \frac{Z_2 Z_0}{Z_2 + Z_0}$$

Therefore,

$$Z_2 \parallel Z_0 = \frac{(j0.20)(j0.10)}{j0.20 + j0.10}$$

$$= j0.0667$$

Hence,

$$I_1 = \frac{E_1}{Z_1 + (Z_2 \parallel Z_0)}$$

$$I_1 = \frac{1}{j0.20 + j0.0667}$$

$$\boxed{I_1 = -j3.75\ pu}$$

The sequence voltage at the fault is

$$V_1 = E_1 - Z_1 I_1$$

$$V_1 = 1 - (j0.20)(-j3.75)$$

$$\boxed{V_1 = 0.25\ pu}$$

For the negative-sequence network,

$$V_2 = -Z_2 I_2$$

At the fault,

$$V_2 = V_1 = 0.25$$

Therefore,

$$I_2 = -\frac{0.25}{j0.20}$$

$$\boxed{I_2 = -j1.25\ pu}$$

Similarly,

$$I_0 = -\frac{0.25}{j0.10}$$

$$\boxed{I_0 = -j2.50\ pu}$$

So,

$$\boxed{I_0 = -j2.50, \quad I_1 = -j3.75, \quad I_2 = -j1.25\ pu}$$

Use the transformation matrix:

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} -j2.50 \\ -j3.75 \\ -j1.25 \end{bmatrix}$$

For phase a,

$$I_a = I_0 + I_1 + I_2$$

$$\boxed{I_a = -j7.50\ pu}$$

Worked Example 1: Three-Phase Symmetrical Fault

SYMMETRICAL FAULT PROBLEM

Problem: A 20 MVA, 10 kV alternator has a subtransient reactance of 5%. Find the additional series external reactance required so that the fault current on a 3-phase short circuit at the terminals does not exceed 8 times full-load current.

Step	Operation	Result
1	Base MVA = 20, Base kV = 10. Calculate base current: Ibase = 20×10^3 / ($\sqrt{3}$ × 10)	Ibase = **1154.7 A**
2	Maximum allowed fault current = 8 × IFL = 8 × 1154.7	If,max = **9237.6 A**
3	Required total pu impedance: Ztotal = 1.0 / (8 × 1.0) (since IFL = 1.0 pu)	Ztotal = **0.125 pu**
4	Alternator reactance already = 5% = 0.05 pu. External reactance needed: Xext = 0.125 − 0.05	Xext = **0.075 pu = 7.5%**
5	Convert to ohms: Xext(Ω) = 0.075 × (10^2/20)	Xext = **0.375 Ω**

Answer: An external series reactance of **0.375 Ω (7.5% on machine base)** limits the 3-phase fault current to exactly 8 × full-load current.